

AYESHA ZAFAR

COMPUTER ENGINEERING STUDENT

NUST , CEME RAWALPINDI | ayeshazafar.n@gmail.com | <https://github.com/nyeshaz1>

PROFESSIONAL SUMMARY

- Developed a Java-based drone simulation module, designed a custom 32-bit RISC processor, and conducted mini-scale research on cognitive load and learning outcomes.
- Proficient in C++, Java, Python, SystemVerilog, embedded systems, microcontroller communication protocols, and research methodologies.
- Collaborative problem-solver with strong project management skills, technical writing ability, and capacity to learn quickly in diverse teams.

PROJECTS

Drone Formation Simulation

Java

- Developed Drone Class for quadcopter dynamics and motion feedback.
- Designed Controller Class with PID/PD thrust navigation.
- Built Environment & Logger Classes for boundaries, data collection, and performance metrics.

Password Strength Checker

C++

- Built Password Strength Checker console application.
- Evaluated length, character types, and missing elements for security feedback.

32-Bit RISC Custom Processor

System Verilog

- Implemented processor supporting I-type, R-type, branch, load, and store instructions.
- Developed core modules: PC, Instruction Memory, Register File, ALU, Data Memory & Control Logic.
- Verified instruction execution and data flow via simulation.

AI Based Micro vs. Macro Explanation : Effects and Results

Research Project

- AI Based Micro vs Macro Explanation : Effects on Cognitive Load and Learning Outcomes in University Students
- Conducted small-scale group research with ~100 university students.
- Ran controlled experiment with 30 participants comparing short vs. detailed explanations and analyzed knowledge retention and comprehension.

WORKSHOP / HANDS-ON EXPERIENCE

- Embedded Systems : Hands-on experience with microcontrollers, I2C, SPI, UART, and IoT protocols.
- Worked on team projects with sensor integration and data communication..

EDUCATION

BE in Computer Engineering

2024 - Present

NUST,CEME Campus, Pakistan

Intermediate in Computer Science (ICS)

2022 - 2024

APSACS, Karachi & Sialkot, Pakistan

Matriculation

2022

FG Minwala Girls Public School, Karachi Cantt, Pakistan

SKILLS & ACHIEVEMENTS

TECHNICAL SKILLS

- C++
- JAVA
- SYSTEM VERILOG
- PYTHON (BASIC)

SOFT SKILLS

- Teamwork
- Problem Solving
- Project Management

Certifications & Leadership

- German Language A1.1
- Campus -Level Archery Competition
- Deputy Dir – Riding & Archery Clubs
- Volunteer – COMPPEC’25 (Promotions , Admin & HR)